

Deferred Outcome Scoring for Scientific Decision Instruments: A Prospective Three-Case Benchmark

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Abstract

Scientific decision instruments often act before the relevant ground truth exists, so contemporaneous agreement cannot itself establish correctness. We define a deferred-outcome benchmark in which an unresolved frontier question, the exact evidence available at decision time, diagnosis and action coordinates, and a later scoring rule are frozen before resolution. Two heterogeneous instruments—an LLM-host-driven research lane and a typed non-LLM controller—then record decisions against the same frozen evidence, and later scientific results score each decision without rewriting the original question. The completed bounded series contains three prospectively valid questions. In all three, the instruments agree on the primary next move, and those moves are later aligned under the frozen scoring maps. Crucially, one case is an explicit counterexample to agreement as validation: both instruments agree on the responsible layer, yet the complete 39,489-instance follow-up supports a different responsibility interpretation while preserving the selected move. A second follow-up exhaustively checks 53 frozen rows outside a sufficient-cone boundary and finds no counterexample, leaving sharpness open rather than converting silence into a theorem. Two earlier candidate questions were retired before scoring when prospectivity became unsafe and remain visible in the audit record. With only three non-random questions from one research programme, we report a case series rather than an agreement rate, reliability coefficient, calibration estimate, or population success probability. The contribution is therefore a temporal evaluation contract and a concrete demonstration that agreement and correctness are distinct scientific objects; broader predictive validity requires a separately frozen larger multi-programme study.

Keywords: research agents; benchmark; deferred outcomes; epistemic control; provenance; scientific workflows.

1 Introduction

When a research system is asked a question whose answer is already known, task success is the natural endpoint. A frontier decision poses a different measurement problem. The system may need to decide which assumption failed, which representation should change, or which experiment should run next before ground truth exists. Repeated sampling from one model can measure self-consistency, but it does not provide an architecturally distinct instrument that can disagree for different reasons.

Q3 defines a benchmark around that gap. The object measured is not whether an LLM is correct and not whether a typed controller is correct. It is whether two different decision procedures, constrained to the same admitted evidence, converge on the same epistemic diagnosis and next action under a protocol frozen before either outcome. Divergence is data, not failure.

The first lane is host-driven and can use an LLM to synthesize a decision, but its evidence-bearing external interactions are captured in content-bound receipts. The second lane uses typed production control modules over a frozen manifest and receipt-transcribed observations; no LLM participates in its decision path. The distinction is architectural rather than absolute: both lanes are constructed inside one programme and share the evidence corpus by design.

Benchmark V0 asks what layer is responsible after a sequence of exact TARE-family closure/refutation results and what move should come next. The result is agreement. The paper reports that measurement and its decision mechanics while resisting the tempting but unsupported next sentence: one aligned instance does not establish instrument reliability.

2 Methods

2.1 Benchmark construct

The V0 protocol freezes a single live frontier question, the admissible receipt corpus, the two lane definitions, scored coordinates, tie/compatibility rules, and the interpretation of divergence before either lane result exists. Scoring separates (1) responsible-layer diagnosis, (2) selected next move, (3) evidence overlap/admissibility, and (4) deferred alignment with later research outcomes.

2.2 Lane A: host-driven instrument

Lane A operates through a generic research harness. Capability requests have deterministic identities derived from canonicalized request content, and results are digest-bound to the exact request. The host may use an LLM for synthesis, but the benchmark answer must be reconstructable from admitted receipt-backed evidence. The lane’s free-text reasoning is therefore not itself an authority source.

2.3 Lane B: typed controller

Lane B receives a frozen campaign manifest containing competing responsibility hypotheses, expected observations, revision mechanics, obligation states, and computation actions. Production epistemic-control modules select a hypothesis and action over typed observations. The decision path contains no LLM. State, decision, and transition records serialize authority grants as false.

2.4 Revision gate

The surviving representation-regime hypothesis is associated with a representation-split revision that requires a frozen regime-membership predicate. At the V0 decision point that obligation is unresolved. Under fail-closed controller semantics the revision is unavailable, leaving characterization computation as the admissible next move.

2.5 Independence and deferred scoring

The lanes are independent in decision procedure but not in evidence source: both use the same committed R6 receipt corpus. Lane B’s manifest was authored/transcribed by a human/LLM process before the native decision, so framing sensitivity remains a possible dependence channel. Deferred scoring uses later research outcomes as a diagnostic alignment signal, not as an externally supplied gold label.

3 Results

3.1 Three prospectively valid frontier questions

The bounded series contains three prospectively frozen scientific questions. The inferential unit is one frontier question, not a receipt, controller state, exact-search row, or host call. Two earlier candidate questions were retired before scoring when later-result-oriented work became visible before instrument freeze; they remain in the audit history but are excluded from the valid series rather than silently replaced after outcome access.

3.2 Case 1: regime characterization

In the first case, Lane A identifies the responsible layer as representation-regime characterization and selects regime-predicate characterization as primary, with a support-two closure test as a complementary probe. Lane B reaches the corresponding typed responsibility state and selects the same primary computational move while withholding a representation revision whose prerequisite obligation is unresolved.

The contemporaneous relation is agreement on the scored diagnosis and primary move. Later exact regime-characterization and support-two results align with the frozen move coordinates. This is one prospective case, not a probability estimate.

3.3 Case 2: outside-cone sharpness remains open

The second case freezes an objective immediately outside a sufficient-cone boundary and asks whether an exact support-two method fails relative to unrestricted dynamic programming. The later analyzer checks 53 frozen rows: three hostile one-object rows, two hostile two-object rows, 24 deterministic random two-object rows and 24 deterministic random three-object rows. It finds zero strict cost gaps in which unrestricted dynamic programming beats the support-two method.

The zero-gap result aligns with both instruments' frozen responsibility and next-move coordinates, but its scientific interpretation is deliberately narrower: it is exact finite-panel evidence and does not enlarge the all-size sufficient cone or prove sharpness.

3.4 Case 3: agreement without diagnostic correctness

The third case changes the objective weights and freezes a complete recomputation over 729 ordered one-object instances and 38,760 reorder-quotiented two-object instances, for 39,489 instances in total. The pre-existing finite-domain predicate has zero mismatches with the reweighted incumbent-exact label on this complete declared domain.

Before that outcome existed, both instruments agreed on a responsibility interpretation tied to an objective-scoped boundary and selected a complete reweighted census as the next move. Under the scoring map frozen before the analyzer/result, the later evidence supports a stronger finite-domain structural-invariance interpretation. Consequently the two instruments *agree with each other* on responsibility while both responsibility scores are false; their selected move nevertheless scores aligned for both lanes.

This is the central counterexample of the series: inter-instrument agreement and later diagnostic correctness are distinct variables. The benchmark therefore does not treat consensus as validation.

3.5 Descriptive series boundary

All three frozen primary moves are later aligned under their case-specific scoring maps, but the questions are non-random and all originate in one research programme. We report the three rows individually. We do not report “100% move accuracy”, a binomial interval, a kappa coefficient, a reliability estimate, calibration, or a generalized success probability.

4 Discussion

The three-case series changes the interpretation of the benchmark. The initial case showed that agreement between heterogeneous scientific decision instruments can be frozen and replayed before a frontier is resolved. The completed series adds the more important observation: agreement can be *wrong at the diagnosis level* even when the agreed next move is useful. The benchmark should therefore be understood as a device for separating contemporaneous decision relations from later scientific scores, not as a consensus mechanism.

This separation matters because the instruments are heterogeneous but not causally independent. Lane A reaches decisions through language-model-assisted evidence synthesis; Lane B applies a typed deterministic controller. Yet both share upstream evidence, programme history and conceptual vocabulary. Their agreement can therefore reflect either convergent reasoning or shared framing. The objective-rescaling case exhibits exactly the failure mode that a consensus interpretation would miss: both lanes select the same diagnostic layer, later evidence supports a different responsibility interpretation, and the selected experiment still turns out to be the right discriminator.

The outside-cone case illustrates a different discipline. An exhaustive zero-gap result over the frozen panel is scientifically informative, but it leaves the all-size sharpness question open. Deferred scoring therefore needs typed scientific outcomes such as aligned, misaligned, unresolved and invalidated rather than a rule that turns every completed computation into validation.

Prospectivity is also part of the measurement rather than administrative metadata. Two candidate questions were retired when outcome-oriented work became visible before instrument freeze. Preserving those contaminated candidates in the audit record, while excluding them from the valid denominator, is preferable to silently substituting easier cases after outcome access.

A larger study could ask whether agreement, disagreement or cannot-check states have predictive value across research programmes. The present paper does not answer that population question. Its bounded result is the temporal evaluation contract and an explicit prospective counterexample showing that agreement is not correctness.

5 Related work and boundary

AgentBench evaluates language-model agents across interactive environments with task-level outcomes [1], and SWE-bench evaluates language models against real software issues with repository-grounded success criteria [2]. Autonomous-science systems such as Coscientist demonstrate tool-using LLM workflows for planning and experimental execution [3]. Provenance research supplies a separate conceptual parent for tracing derived decisions to source records [4]. These works support the neighboring benchmark, agent, and provenance contexts; none is cited as evidence that agreement between two instruments is itself predictive of scientific correctness.

Q3’s bounded candidate residual is the specific measurement contract: the same frozen frontier question, architecturally distinct decision paths, receipt-bound evidence, divergence as an admis-

sible outcome, and deferred scoring when contemporaneous ground truth is unavailable. That residual remains conditional on a fresh submission-date search; the manuscript therefore does not claim that no prior benchmark measures any form of cross-instrument agreement.

The broader host/capability receipt architecture is not claimed as Q3’s standalone systems novelty. That subject is routed to P15, which requires its own donor matrix and protocol before becoming a systems-paper claim.

6 Limitations

The decisive limitation is still sample size, but it is no longer a missing-data blocker for the bounded paper: the current evidence is a three-question prospective case series. Three non-random questions do not authorize a reliability estimate, agreement rate, kappa coefficient, calibration claim, binomial success probability or population generalization. The stronger predictive-validity question remains a separately frozen successor requiring a larger multi-programme registry.

The lanes are not fully independent. They share the receipt corpus, originate in the same research programme, and Lane B’s manifest is human/LLM-authored even though the native decision is not. Manifest construction can encode framing choices. Lane A’s language-model-assisted synthesis may also share conceptual priors with the people who designed the typed hypotheses. Architectural heterogeneity is therefore not independent-investigator authority.

Deferred scoring is endogenous to the same programme rather than external ground truth. The third case shows why that distinction matters: both instruments agree on a diagnosis that later programme evidence scores as wrong while their selected move remains aligned. This supplies a prospective counterexample to agreement-as-validation, not an estimate of how often agreement is wrong.

The exact finite-panel results are also scoped. Zero gaps across 53 outside-cone rows do not prove an all-size sharpness theorem, and zero predicate mismatches across the complete declared one- and two-object domain do not establish an all-size weighted law.

Finally, reproducible receipts do not make a scientific decision correct. They make the evidence and decision path attributable. The benchmark’s bounded contribution is temporal measurement with preserved contamination and deferred outcomes; predictive validity, cross-programme generality and external scientific adjudication remain open.

7 Conclusion

Deferred-outcome scoring provides a measurable object for live scientific decisions that do not yet have a gold label. Across three prospectively valid questions, an LLM-host-driven research lane and a typed non-LLM controller record decisions before later scientific outcomes exist, and the frozen scoring maps preserve agreement, misalignment and unresolved scientific scope without rewriting the original questions.

The most important result is negative for consensus-based validation: in one complete finite-domain follow-up, the two instruments agree on the responsible layer and both diagnoses are later scored as wrong, even though their agreed next move remains aligned. Agreement is therefore an observation about the instruments, not scientific authority.

The bounded contribution is complete at the level claimed here: a prospective temporal benchmark contract, three scored cases, preserved contaminated candidates and an explicit agreement-not-correctness counterexample. A larger multi-programme successor is needed only for reliability,

calibration or predictive-validity claims; it is not a prerequisite for publishing this bounded benchmark paper.

8 Reproducibility

The V0 protocol, lane receipts, result JSON, and workspace archive are committed under `development/orion-q-max-r0/`. Lane A can be audited by recomputing request/result identities and tracing the final synthesis to its verified evidence items. Lane B can be replayed from the frozen manifest and receipt-transcribed observations; a manifest-digest mismatch must fail before a scientific decision is accepted.

Deferred scoring is independently reconstructable from the committed R6P/R6Q result artifacts. Historical invalid-content recovery should be tested with `packages/orion-research-harness/tests/test_invalid_content_recovery.py`.

The root `REPRODUCE.md` gives the intended verification order.

The missing benchmark series is not a reproducibility failure. It is missing evidence and requires new pre-outcome protocols/executions.

9 Ethics, safety and resource statement

No human participants, animals, clinical data, or personal data are used. The main integrity risks are authority laundering, hidden evidence asymmetry between lanes, post-outcome scoring changes, and selective reporting of agreement. The protocol addresses these by freezing evidence/scoring before outcomes, making divergence an admissible terminal, and recording all authority grants as false in the typed lane.

The benchmark should not be used to claim that consensus is correctness. Future series must retain disagreements and adverse deferred outcomes. Receipt coverage is provenance, not evidence quality, and neither lane is permitted to turn a host/tool failure into scientific evidence.

References

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